



LIANGSHAN ENTERPRISE

PHYSICAL SECURITY PLAN

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ix. Physical Security Plan

1. Introduction

Liangshan Enterprises, a global leader in architectural innovation, recognizes the importance of a robust physical security plan to protect its infrastructure, personnel, and sensitive assets. This plan outlines a comprehensive framework addressing access control, surveillance, emergency response, and maintenance to ensure the safety and operational continuity of the enterprise.

2. Contact List

The following table lists key personnel from the physical security team responsible for implementing and managing Liangshan Enterprises' physical security plan. The team ensures the safety and security of employees, clients, and infrastructure across all facilities, with a primary focus on the headquarters in Beijing.

Name	Title/Role	Location	Phone Number	Email
Jian Wei	Chief Security Officer (CSO)	Beijing (HQ)	+86-10-3456-7890	jwei@liangshan.com
Mei Ling	Security Operations Manager	Beijing (HQ)	+86-10-4567-8901	mling@liangshan.com
Yao Zhang	Head of Physical Security	Beijing (HQ)	+86-10-5678-2345	yzhang@liangshan.com
Ethan Grant	Surveillance Team Lead	Beijing (HQ)	+86-10-6789-1234	egrant@liangshan.com
Aisha Khan	Perimeter Security Supervisor	London Office	+44-20-3345-6789	akhan@liangshan.com
Hange Zoe	Physical Security Specialist	New York Office	+1-212-456-7890	hzoe@liangshan.com

Omar Hassan	Access Control Manager	Dubai Office	+971-4-5678-2345	ohassan@liangshan.com
Sophia Li	Fire and Emergency Response	Beijing (HQ)	+86-10-7890-5678	sli@liangshan.com
Ming Zhao	Maintenance Supervisor	Beijing (HQ)	+86-10-9876-5432	mzhao@liangshan.com
Emily Carter	Vendor Liaison Specialist	Beijing (HQ)	+86-10-1234-5678	ecarter@liangshan.com
Li Hua	Fire Warden	Beijing (HQ)	+86-10-4567-8902	lhua@liangshan.com
Billy Han	Emergency Vendor Contact	London Office	+44-20-3456-7891	bHan@liangshan.com
Omar Khalid	HVAC Maintenance Technician	Dubai Office	+971-4-2345-6781	okhalid@liangshan.com

Table 9.1: Physical Security Team Contact Information

The HQ's physical security team is divided into specialized sub-teams to manage distinct aspects of the security plan, including perimeter security, surveillance, access control, and emergency response.

2.1 Overview of Beijing HQ Team

- **Access Control:**
 - Managed by **Mei Ling (Security Operations Manager)**.
 - Role-based biometric systems and card access are configured to allow access only to authorized personnel.
 - Visitor management system is supervised by the **Access Control Desk** under **Ethan Grant**.
- **Surveillance Team:**
 - Oversees 24/7 monitoring of CCTV feeds across the HQ, ensuring no blind spots.
 - Led by **Ethan Grant**, supported by four surveillance operators.

- **Emergency Response Team:**
 - Led by **Sophia Li**, responsible for coordinating evacuation drills, fire safety training, and emergency protocol reviews.

2.2 Overview of Regional Office (London) Team

- **Security Management:**
 - Managed by **Aisha Khan (Perimeter Security Supervisor)**.
 - Perimeter monitored using AI-driven surveillance cameras and motion detection systems.
 - Regional SOC team coordinates responses for anomalies or breaches.

2.3 Overview of Regional Office (New York) Team

- **Access Control and Surveillance:**
 - Managed by **Hange Zoe (Physical Security Specialist)**.
 - Dual-layer access control with RFID-based entry for employees and biometric systems for restricted areas.

2.4 Overview of Regional Office (Dubai) Team

- **Emergency Preparedness and Perimeter Security:**
 - Managed by **Omar Hassan (Access Control Manager)**.
 - Focuses on emergency evacuation plans and securing access points with anti-climbing walls and smart locks.

3 Building Security

3.1 Access Control

- **Biometric Authentication:** Installed at all sensitive areas, including executive offices, R&D labs, and data centers.
- **RFID Card Access:** Issued to employees based on roles, with limited access to high-security areas. Lost or stolen cards are immediately deactivated, and temporary replacements are issued under strict verification.
- **Employee Training:** Employees are educated on preventing tailgating and the risks associated with misplaced credentials.
- **Visitor Management:**
 - Visitors are required to provide government-issued IDs, which are scanned and logged into the visitor management system.
 - Temporary RFID badges are issued, configured to grant access to only required areas.

Visitors are always accompanied by authorized personnel and monitored through surveillance cameras during their stay.

3.2 Security Personnel

- **Hiring Criteria:**
 - Background checks, qualifications, and physical fitness are mandatory.
 - Physical fitness tests and psychological evaluations are mandatory to ensure readiness for high-stress situations.
 - Training programs include emergency response protocol, access control technologies and surveillance systems, and scenario-based conflict resolution exercises.
- **Deployment:**
 - Guards are stationed at critical points such as facility entrances, parking lots, server rooms, and executive offices.
 - Mobile patrol units monitor internal and external zones, ensuring no blind spots.
 - A rotating shift system ensures uninterrupted 24/7 coverage.
 - Guard performance and alertness are reviewed monthly by the Security Manager.

3.3 Locks and Keys

- **Secure Access:**
 - Biometric locks are installed at critical areas, including data centers, R&D labs, and executive suites, requiring fingerprint or retinal scans for access. Dual-authentication systems (ID card and biometric verification) are in place for highly sensitive zones.
 - Electronic locks default to a locked state during power outages, ensuring continued restriction of unauthorized access.
- **Key Management:**
 - Physical keys are only issued for non-critical areas such as storage rooms or general office spaces.
 - Key issuance is logged, and individuals are required to sign receipts for accountability.
 - A centralized key management system tracks all issued keys, logging usage history and active holders.
 - Keys for critical areas are stored in secure cabinets with biometric access.

4. Perimeter Security

4.1 Walls, Fencing, and Gates

- **Perimeter Walls:**
 - All Liangshan facilities are surrounded by reinforced concrete walls at least 10 feet high, equipped with anti-climbing measures like barbed wire and electric fencing. In high-risk zones, motion sensors are integrated into the fencing to detect breaches.
- **Entry Gates:**
 - Automated entry gates are equipped with RFID-enabled access control systems, allowing seamless and secure entry for authorized vehicles and personnel. Security guards stationed at entry points verify credentials and monitor vehicle inspections.
- **Monitoring and Maintenance:**
 - Regular inspections for cracks, wear, or tampering on walls, gates, and fencing.
 - Vulnerabilities or damage are repaired within 24 hours of identification.
 - Scheduled weekly inspections by the Physical Security Team.

5. Surveillance and Monitoring

5.1 CCTV Systems

- **Coverage:**
 - Strategically placed cameras provide 360-degree coverage of perimeter walls, parking areas, entry/exit points, and internal corridors.
 - High-security zones such as data centers are equipped with cameras featuring AI-powered anomaly detection.
- **Monitoring:**
 - All camera feeds are monitored 24/7 by the Physical Security Team, based in the headquarters and regional office security rooms.
- **Maintenance:**
 - Cameras are inspected biweekly and tested monthly to ensure uninterrupted operation.
- **Backup Systems:**
 - A secondary monitoring system is maintained on a cloud-based platform, enabling remote access for redundancy in case of SOC system failures.

- On-site storage has redundant storage devices, and footage is automatically backed up every 24 hours.
- **Action Plan for Failures:**
 - In case of camera or monitoring system failure, priority zones (data centers, executive offices, and R&D labs) are immediately patrolled by security personnel.
 - Downtime logs are maintained, and a troubleshooting protocol is initiated by the IT and maintenance teams within one hour.

5.2 Alarms and Sensors

- **Intrusion Detection:**
 - Motion detectors installed on doors, windows, and perimeter walls trigger alarms upon detecting unauthorized activity.
- **Environmental Sensors:**
 - Smoke, fire, and water leakage sensors are installed in server rooms, data centers, and kitchens to mitigate potential hazards.
- **Integration:**
 - All alarms are directly connected to the Physical Security Team, ensuring immediate alerts and coordinated responses.

6. Facility-Specific Measures

6.1 Headquarters (Beijing)

- **Access Control:**
 - Biometric authentication is required for all critical zones, including SOC, executive offices, and R&D labs.
- **Elevator Security:**
 - Role-based access controls limit movement between floors.
- **Parking Security:**
 - RFID barriers and CCTV surveillance integrated with motion-activated lighting ensure vehicle safety.

6.2 Regional Offices (London, New York, Dubai)

- **Entry Protocols:**
 - The Regional Physical Security Team oversees access permissions, visitor management, and incident reporting.
- **Emergency Exits:**
 - Clearly marked with backup lighting, equipped with panic bars for easy egress.

6.3 Design Studios (Tokyo, Paris)

- **Enhanced Access Controls:**
 - Restricted entry to authorized employees and clients.
 - Biometric authentication for visualization labs and 3D rendering zones.
- **IoT Integration:**
 - IoT sensors monitor temperature, humidity, and equipment usage to protect high-value assets.
- **Collaboration with Local Authorities:**
 - Memorandums of Understanding (MOUs) are signed with local fire and police departments, detailing response protocols and access permissions.
- **Elevator Security:**
 - Role-based access controls for elevators restrict access to sensitive floors, such as visualization labs or executive meeting rooms.

6.4 Data Centers (Beijing, London)

- **Perimeter Security:**
 - Multiple layers of fencing, monitored by surveillance cameras with AI-powered analytics.
- **Internal Security:**
 - Dual-authentication systems (RFID + biometrics) for entry into server and storage zones.
- **Local Response Teams:**
 - Regular communication is maintained with local first responders, who are provided with detailed floor plans and emergency contacts.

7. Fire Safety Measures

7.1 Detection Systems

- **Thermal Sensors:**
 - Advanced thermal detection systems are installed in high-risk areas such as server rooms, kitchens, and electrical closets.
 - These sensors detect rapid temperature increases, often indicative of fire, even before visible flames or smoke are present.
 - Integrated with the central monitoring system to alert the Physical Security Team for immediate response.

- **Smoke Alarms:**

- Networked smoke detectors are installed across all facilities, including hallways, common areas, offices, and storage spaces.
- Alarms are interconnected and designed to provide synchronized alerts, ensuring comprehensive coverage.
- Notifications are sent instantly to the Physical Security Team, allowing quick coordination with local fire services.
- Regular maintenance schedules include cleaning and battery replacements to prevent false alarms or non-functioning units.

7.2 Suppression Systems

- **Sprinkler Systems:**

- Water-based sprinkler systems are strategically installed in common areas, such as hallways, cafeterias, parking lots, and workspaces.
- Sprinklers are configured to activate locally in affected areas, minimizing water damage to unaffected zones.
- Regular inspections ensure no clogging or malfunctions.

- **Gas Suppression Systems:**

- Non-water-based fire suppression systems are installed in critical areas such as server rooms, data centers, and IT labs.
- These systems extinguish fires without causing damage to sensitive electronic equipment.
- Manual override controls are available for emergency use by trained personnel.
- Warning systems are installed to ensure the evacuation of personnel before gas discharge.

7.3 Evacuation Protocols

- **Marked Exit Routes and Assembly Points:**

- Clearly marked and illuminated emergency exit routes are present on all floors, leading to safe outdoor assembly points.
- Exit signs are equipped with backup batteries to remain operational during power outages.
- Assembly points are located a safe distance from the building to prevent harm from fire, smoke, or structural collapse.

- **Emergency Exit Infrastructure:**

- Emergency doors are equipped with panic bars for quick and unimpeded evacuation.
- Doors automatically unlock during emergencies, ensuring no delays in egress.

- Staircases are designated as primary evacuation routes, with elevators programmed to halt during emergencies for safety.
- **Biannual Fire Drills:**
 - Fire drills are conducted twice a year to familiarize employees with evacuation protocols and designated assembly points.
 - Drills are monitored and assessed for efficiency, with feedback provided to improve response times.
 - Participation is mandatory for all employees, including new hires, ensuring awareness at all organizational levels.
- **Fire Wardens:**
 - Designated fire wardens are assigned for each floor to assist with evacuations, ensuring no one is left behind.
 - Wardens are equipped with emergency radios for direct communication with the Physical Security Team and first responders.
- **Collaboration with Local Fire Departments:**
 - Pre-established coordination with local fire services ensures swift response during fire incidents.
 - Site-specific fire safety plans are shared with local fire departments, providing them with essential details, such as access points, hazardous material locations, and water sources.

8. Utility and Structural Security

8.1 Power Management

- **Uninterruptible Power Supplies (UPS):**
 - High-capacity UPS systems are deployed to provide immediate power backup for all critical systems, including servers, data centers, and essential workstations.
 - UPS units are equipped with battery health monitoring systems to identify performance issues proactively.
 - Annual stress tests simulate prolonged outages to ensure UPS reliability and efficiency.
- **Backup Generators:**
 - Generators with sufficient capacity are installed to sustain operations during extended power outages.
 - Fuel reserves are regularly checked, and generators undergo a monthly load test to verify readiness.
 - Generators are housed in secure, weather-resistant enclosures to protect them from tampering and environmental damage.

- Automatic transfer switches (ATS) ensure seamless transition between main power and backup systems.
- **Extended Blackouts:**
 - Additional portable generators are maintained at each site to handle blackouts lasting over 48 hours.
 - A fuel resupply agreement with local providers ensures continuous operation during extended outages.

8.2 HVAC Systems

- **Regular Maintenance:**
 - HVAC systems across all facilities are inspected and serviced quarterly to ensure consistent performance and energy efficiency.
 - Maintenance includes cleaning filters, checking refrigerant levels, and calibrating thermostats to avoid system breakdowns.
 - Technicians are trained to identify early signs of wear and tear, reducing downtime risks.
- **Environmental Monitoring in Server Rooms:**
 - Precision cooling systems are installed in server rooms and data centers to maintain optimal operating temperatures.
 - IoT-enabled environmental sensors continuously monitor temperature, humidity, and airflow, triggering alerts in case of deviations.
 - Emergency cooling measures, such as portable air conditioning units, are available to mitigate risks during system malfunctions.

8.3 Water Leak Management

- **Water Sensors:**
 - Advanced water leakage sensors are strategically placed near restrooms, kitchens, and water pipelines to detect and mitigate risks early.
 - Sensors are integrated with the central monitoring system, enabling real-time alerts to the Physical Security Team for immediate action.
- **Regular Inspections:**
 - Facility maintenance teams conduct monthly inspections of plumbing systems, including water tanks and pipelines, to identify vulnerabilities.
 - Drip pans are installed beneath critical water lines to prevent water damage in the event of minor leaks.
- **Emergency Shutoff Valves:**

- Automated shutoff valves are installed in critical zones to halt water flow during emergencies, minimizing potential damage.
- Valves are accessible to maintenance staff and connected to the monitoring system for remote activation.
- **Emergency Response for Significant Leaks:**
 - Emergency repair kits (sealants, pumps) are available on-site to manage large leaks before external technicians arrive.
 - Physical Security Team coordinates with building maintenance staff to shut down affected water lines within 15 minutes of detection.

9. Inventory Management

Inventory management is a cornerstone of operational security, ensuring that Liangshan's assets and sensitive documents are tracked, protected, and readily accessible during critical operations.

9.1 Equipment Tracking

- **Tagging and Labeling:** All IT and office equipment, including laptops, servers, and peripherals, are tagged with unique identifiers for tracking purposes.
- **Inventory Audits:** Quarterly audits are conducted to reconcile inventory records with physical assets, ensuring accountability and accuracy.
- **Real-Time Tracking:** RFID and barcode scanning systems streamline inventory tracking, enabling efficient updates to the central inventory database.
- **Loss Prevention:** Discrepancies in equipment records trigger investigations to prevent theft or unauthorized usage.

9.2 Sensitive Documents

- **Fireproof Cabinets:**
 - Critical documents, such as client contracts, blueprints, and financial records, are stored in fire-resistant cabinets located in secure zones.
 - Cabinets are equipped with biometric locks to ensure access is restricted to authorized personnel only.
- **Digital Backups:**
 - Sensitive documents are digitized, encrypted, and stored in secure data centers compliant with ISO 27001 standards.
 - Backups are updated weekly, ensuring the latest versions of all critical documents are available.
 - Multi-factor authentication (MFA) is required to access digital backups, adding an additional layer of security.

- **Access Control:**
 - Document access logs are maintained to track who accessed specific files and when.
 - Unauthorized access attempts are flagged and reported to the Physical Security Team for review.

10. Maintenance and Audits

To ensure the effectiveness and adaptability of Liangshan Enterprise's physical security measures, a structured approach to maintenance and audits is crucial. This involves regular inspections, systematic reviews, and proactive feedback mechanisms.

10.1 Routine Inspections

- **Perimeter Security:**
 - Weekly inspections of perimeter walls, fencing, and gates are conducted to identify potential breaches, wear, or damage.
 - Any issues, such as broken barbed wire or faulty electric fencing, are immediately logged and repaired within 24 hours.
- **Locks and Access Control Systems:**
 - All biometric locks, RFID systems, and manual locks are inspected weekly to ensure they are functioning correctly.
 - A monthly review of key distribution records ensures that access permissions remain aligned with employee roles and responsibilities.
- **Surveillance Systems:**
 - Weekly operational checks of surveillance cameras, including pan-tilt-zoom (PTZ) functions, focus, and recording quality.
 - Monthly audits of CCTV footage logs ensure coverage of critical areas and identify potential blind spots.
- **Visitor Records and Logs:**
 - A monthly audit of visitor records, access logs, and badge issuance ensures compliance with access policies.
 - Anomalies, such as unverified visitors or unreturned temporary badges, are flagged for investigation.

10.2 Continuous Improvement

- **Feedback Loops:**
 - Employees are encouraged to report vulnerabilities or concerns through an anonymous feedback system managed by the Physical Security Team.

- Suggestions are reviewed monthly, and actionable items are incorporated into security updates.
- **Annual Security Reviews:**
 - Comprehensive annual reviews of all physical security measures are conducted to assess performance and identify areas for improvement.
 - The reviews incorporate insights from industry benchmarks, evolving threat landscapes, and past incident reports.
- **Implementation of New Technologies:**
 - Emerging security technologies, such as AI-driven surveillance or advanced intrusion detection systems, are evaluated annually for integration into the existing framework.

11. Employee Training and Awareness

Training and awareness programs are integral to Liangshan Enterprise's physical security strategy, ensuring that all personnel understand their roles in maintaining a secure environment.

11.1 Training Programs

- **Workshops on Physical Security Protocols:**
 - Quarterly workshops educate employees on best practices for physical security, such as properly securing workstations, recognizing unauthorized personnel, and reporting suspicious activities.
 - Department-specific sessions address unique security considerations for areas like data centers, executive offices, and R&D labs.
- **Emergency Evacuation Training:**
 - Employees participate in biannual sessions on evacuation procedures, including recognizing emergency exits, understanding assembly point locations, and following evacuation routes.
 - Managers receive additional training on leading teams during evacuations and accounting for personnel.
- **Fire Safety and Hazard Management:**
 - Regular fire safety workshops focus on using fire extinguishers, recognizing fire hazards, and understanding alarm systems.
 - Employees are taught how to respond calmly and effectively during fire emergencies.
- **Preventing Tailgating and Workspace Security:**
 - Awareness sessions teach employees how to recognize and prevent tailgating incidents, emphasizing vigilance during badge scans.

- Employees learn to secure their workspaces by locking devices, storing sensitive documents securely, and maintaining a clutter-free environment.
- **Advanced Threat Scenarios:**
 - Employees responsible for outdoor security receive training on detecting and mitigating unauthorized drone activity, including reporting and neutralizing potential threats.
- **Cybersecurity Awareness:**
 - Basic modules on recognizing phishing attempts, avoiding social engineering, and secure password management are incorporated.

11.2 Simulations

- **Intrusion and Breach Drills:**
 - Quarterly drills simulate potential intrusion scenarios, such as unauthorized entry attempts or breached perimeters.
 - Employees are evaluated on their responses, including notifying the Physical Security Team and following lockdown procedures.
- **Natural Disaster Simulations:**
 - Scenarios for earthquakes, floods, or severe weather conditions are simulated semi-annually.
 - Employees practice safe evacuation routes and receive guidance on protecting critical assets during such events.
- **Post-Drill Reviews:**
 - After each simulation, detailed feedback sessions identify strengths and areas for improvement in employee performance and response protocols.
 - Actionable recommendations from these reviews are incorporated into future training sessions.

12. Case 01: Unauthorized Access Attempt at Liangshan Enterprises Headquarters

12.1 Incident Overview

On April 12, 2024, at approximately 10:45 PM, an unauthorized access attempt was detected at Liangshan Enterprises' headquarters in Beijing. The incident involved a vehicle attempting to bypass security at the main entry gate and an individual attempting to climb the perimeter wall near the western boundary. The Physical Security Team responded promptly, leveraging the measures outlined in Liangshan's Physical Security Plan (PSP) to contain the threat and prevent any breach.

12.2 Timeline of Events

- **10:45 PM:**
 - Motion sensors on the western perimeter wall detected unusual activity.
 - Simultaneously, an unauthorized vehicle attempted to enter the premises by tailgating an employee's car at the automated RFID-enabled gate.
- **10:46 PM:**
 - The CCTV system, integrated with AI-powered anomaly detection, flagged the suspicious activity.
 - Alerts were immediately sent to the Physical Security Team via the central monitoring system.
- **10:47 PM:**
 - The western perimeter camera feed confirmed an individual attempting to scale the wall using a rope.
 - At the entry gate, the surveillance team identified that the unauthorized vehicle had a falsified RFID sticker.
- **10:48 PM:**
 - Security personnel stationed at the gate stopped the vehicle and requested identification. When the driver failed to produce valid credentials, the gate guards activated a lockdown protocol, isolating the vehicle within the secure entry bay.
 - A mobile patrol unit was dispatched to the western perimeter to intercept the individual attempting to climb the wall.
- **10:50 PM:**
 - Guards apprehended the intruder at the wall. The individual, who claimed to be delivering a package, could not provide a valid delivery request or identification.

- **10:55 PM:**
 - The vehicle was inspected, revealing surveillance equipment and tools for bypassing security systems.
 - Local law enforcement was contacted for immediate assistance, as per the emergency protocol outlined in the PSP.
- **11:30 PM:**
 - Both the individual and the vehicle were handed over to law enforcement authorities. A detailed incident report was compiled by the Physical Security Team.

12.3 Analysis of PSP Implementation

12.3.1 Detection Systems:

- **Motion Sensors:** The motion detectors installed on the western perimeter wall effectively identified the climbing attempt in real-time.
- **CCTV System with AI Analytics:** Cameras equipped with AI-powered anomaly detection successfully flagged the unauthorized activities at both the wall and the entry gate.
- **Environmental Monitoring:** While not directly triggered during this incident, the integration of smoke and water sensors in other critical areas ensured that no additional risks were overlooked during the response.

12.3.2 Access Control:

- **Automated Entry Gate with RFID Authentication:** The falsified RFID sticker was quickly identified by the gate's system, preventing unauthorized access.
- **Visitor Management Protocols:** The failure of the driver to provide identification ensured that the security personnel adhered to strict access policies.

12.3.3 Security Personnel:

- Guards were positioned strategically at entry points and key zones, enabling immediate intervention.
- The mobile patrol unit reached the western perimeter wall within three minutes, demonstrating the efficiency of deployment protocols.

12.3.4 Surveillance and Monitoring:

- The 24/7 monitoring of CCTV feeds allowed the Physical Security Team to respond promptly to the dual threat.
- The backup system ensured uninterrupted video recording, which was later provided to law enforcement for investigation.

12.3.5 Emergency Response Coordination:

- The Physical Security Team's immediate communication with local law enforcement ensured that the suspects were apprehended and that no additional threats persisted.
- The emergency response drills conducted biannually ensured that all team members executed their roles effectively during the incident.

12.4 Post-Incident Actions

12.4.1 Incident Review and Feedback:

- A review meeting was conducted the next morning, led by Jian Wei (Chief Security Officer), to evaluate the effectiveness of the response.
- Feedback highlighted the rapid response and adherence to protocols but identified minor areas for improvement, such as faster dispatching of patrol units.

12.4.2 Updates to the Physical Security Plan:

- **Enhanced Training:** Security personnel received additional training on recognizing falsified RFID credentials.
- **Perimeter Reinforcement:** Motion sensors on the western wall were recalibrated for higher sensitivity, and additional cameras were installed to cover previously unmonitored angles.
- **Visitor Policies:** Vendor verification procedures were reviewed and updated to ensure packages and personnel are pre-approved before arrival.

12.4.3 Drills and Simulations:

- An intrusion simulation was conducted one month later to test the updated protocols, ensuring all improvements were effectively implemented.